Does the Voice Change the Grade? A Causal Audit of Voice Effects in Audio-LLM Judges, with a Cautionary Replication

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Abstract

Audio language models now grade spoken answers directly: automated oral exams, AI interview screening, and speech leaderboards all use an audio-input model as the judge, and all of them assume the judge scores what was said, not how it sounded. We test that assumption three ways: a text-to-speech audit that holds 24 interview answers fixed while varying accent and gender (192 clips, three judges from two vendor families), a replication battery that re-renders the same design with six additional voices and a decomposed rubric, and a real-speech arm built from L2-ARCTIC (24 speakers, six first languages) against matched native recordings. The audit initially produced a headline: Gemini 3.1 Pro scored Indian-accented answers 2.5 points lower than the identical American-accented answer ($p = 0.005$), localized to the acoustic channel by an accent-by-grading-mode interaction (-6.04 versus own-transcript grading, $p = 4 \times 10^{-4}$). The replication battery dissolved it. Across six additional voices the Indian shift is -0.76 ($p = 0.35$), with per-voice swings from $+2.08$ to -4.58: the voice, not the accent, moves the score. Under a decomposed four-dimension rubric on the original voices the penalty vanishes entirely (all dimensions n.s.). We report this as our pre-registered discipline requires, and we flag the general lesson: single-voice, single-rubric audit designs, the norm in this literature, can manufacture an accent-bias finding. On real speech the picture inverts: both Gemini judges penalise every non-native L1, from -4.3 (Hindi) to -15.7 (Mandarin) points under a read-aloud rubric, all $p \leq 4 \times 10^{-4}$, with close cross-judge agreement, while the open-weight Qwen2-Audio-7B barely reacts on its compressed scale. Twelve of these real-speech cells plus four TTS cells survive Benjamini-Hochberg correction across all 33 accent-judge-arm tests. What an audio judge measures depends on the voice sample, the rubric, and the judge family; an audit that fixes all three can say more about its own design than about the model.

1 Introduction

An audio language model that grades a spoken answer is being asked to do two things at once: understand the content, and ignore everything about the voice that carries it. Three deployed systems already depend on that second, unstated assumption holding. Automated oral-language assessment tools grade non-native speakers on scales that claim to measure content, not accent (Ma et al., 2025). AI interview-screening pipelines already process hundreds of thousands of real candidate interviews with an LLM-as-judge step in the loop (Allbert et al., 2025), on top of a pre-LLM fairness literature that already catalogued bias sources in automated video interviews (Booth et al., 2023). Speech leaderboards increasingly use an audio-capable model, rather than a human panel, to rank system outputs (Manakul et al., 2025). None of these deployments has been checked against the one manipulation that would prove or disprove the assumption directly: holding what is said perfectly fixed, and changing only who appears to be saying it.

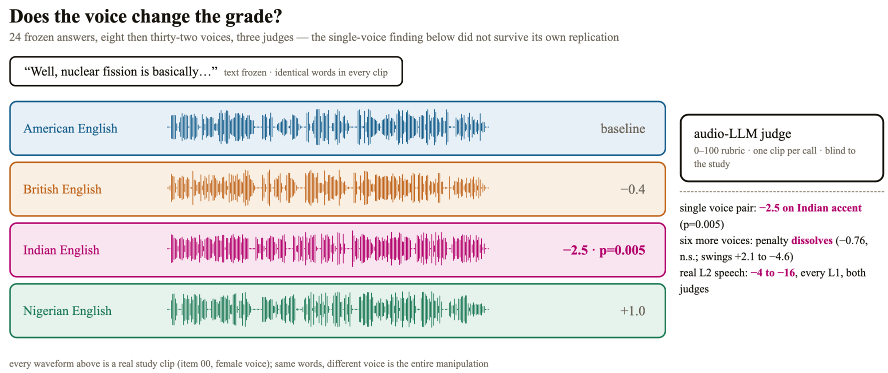

Figure 1: Study design (single-voice TTS arm). One frozen spoken answer, rendered in four accents and two genders by TTS, scored blind by audio-input judges. Any score gap between renderings of the same item is a voice effect by construction. The −2.5 Indian-accent cell previewed at right is the single-voice arm's finding; §4.2 shows it does not survive replication across voices.

We ran that check, and this paper reports what happened when we then subjected our own result to the replication battery a sceptical reviewer would demand. The first arm authors 24 short answers once, freezes the text, and renders each answer in four accents and two speaker genders with a single text-to-speech voice pair, the standard design in this literature. It produced a clean, significant, channel-localized accent penalty under one judge. The second arm re-rendered the same design with six additional voices and a decomposed rubric, and the penalty dissolved: what looked like an accent effect was a property of one voice pair under one rubric shape. The third arm replaced synthesis with real speech, 24 L2-ARCTIC speakers against matched native recordings, where both Gemini judges penalise every non-native first language, heavily, consistently, and in close agreement with each other.

This paper makes five contributions.

1. A three-arm causal audit design, single-voice TTS, multi-voice TTS replication, and real speech, that puts an audio-input model in the grader role with the evaluee's voice as the manipulated variable (Section 3).
2. A cautionary methods finding: a single-voice, single-rubric audit produced a significant, multiple-comparison-surviving, channel-localized "accent bias" that did not survive voice replication or rubric decomposition (Section 4.2). Single-voice designs are the norm in the work we review; the failure mode generalizes.
3. A decomposition design with explicit interaction tests that localizes the (voice-level) penalty to the acoustic channel: grading the same audio from a transcript removes it (−6.04 interaction, $p = 4 \times 10^{-4}$; Section 4.3).
4. A real-speech arm showing large, cross-judge-consistent penalties for all six non-native L1s under a read-aloud rubric, with a stable per-L1 gradient, alongside a third judge family that barely reacts (Section 4.4).
5. A delivery-perturbation battery, filled pauses, speaking rate, telephone-bandwidth audio, and background noise, tested under both a neutral rubric and an explicit ignore-delivery instruction, null at this sample size (Section 4.5).

2 What Was Known and What Is New

The premise that audio-language-model judges carry biases is not new. What is new is which bias, in which role, and what a replication battery does to it.

AudioJudge (Manakul et al., 2025) establishes that large audio models can act as judges of speech, and catalogs verbosity and position bias in that role. It never varies the identity of the speaker being judged; its evaluees are text-to-speech systems, not people.

The Voice Behind the Words (Satish et al., 2026a), and its companion study using the same cohort and an explicitly judge-free protocol (Satish et al., 2026b), are the closest papers by topic. It varies the accent and gender of a question-asker across 2,880 controlled interactions and finds that a SpeechLLM's own reply is rated less helpful for some accents. But the judge in that paper is explicitly text-only and voice-blind: it reads only the assistant's written reply, never the audio. It cannot show, and does not claim, that an audio-input judge's score of an evaluee's own answer moves with that evaluee's voice, which is the question this paper asks.

BiasInEar (Wei et al., 2026) varies the voice of a spoken multiple-choice question and measures whether the model's own answer choice changes. The model there is the test-taker, not a judge. ParaPairAudioBench (Jeon et al., 2026) puts an audio model in the judge role and tests whether it can detect paralinguistic differences when told to look for them, the inverse of asking whether they leak into an unrelated content score unprompted. Bias Analysis of L2 Speaking Assessment Systems (Labroo et al., 2026) audits a real speaking-assessment grader for encoded speaker attributes, but representationally, on naturally varying speakers, not through a controlled same-content voice swap.

What remained uncovered, and is the subject of this paper, is the exact intersection: an audio-input model in the grader role, the evaluee's own voice identity as a randomized treatment, and a content score as the outcome, with the text held identical across every condition.

Three adjacent clusters do not close the gap: judge-construction work without a bias axis (Wang et al., 2025; Zhang et al., 2026; Chen et al., 2025; Park et al., 2026; Chiang et al., 2025; Liu et al., 2025; Takagi et al., 2026); advisor-role work measuring the model's own response, not a grade (Wu et al., 2025; Lin et al., 2026; Satish et al., 2025b;a; Choi et al., 2025); and observational audits of deployed L2 grading (Uehara, 2026; Ginjala et al., 2026; Koenecke et al., 2020). Miller et al. (2026) comes closest procedurally, holding a transcript fixed while varying affect to test evidence use, but manipulates delivery, not demographic identity. TTS-instrument validity work (Zhong et al., 2025; Huang et al., 2026; Xinyuan et al., 2025; Yang et al., 2025; Ren et al., 2026) informs our manipulation checks; our multivoice result (§4.2) is itself evidence for that literature's central worry. Wang and Sun (2026) uses the same real-speech corpus we use as our causal anchor, for a diagnostic-feedback task rather than content grading. The human-listener precedent predates audio language models: Kang and Rubin (2009) show identical recorded speech is rated less comprehensible under a different apparent speaker identity, the causal logic we apply to an artificial judge.

Significance. If a judge's score of a spoken answer depends on the speaker's voice, every deployed pipeline above is partly scoring the speaker rather than the answer. Our results sharpen this in two directions at once: audits using one synthetic voice per condition can overstate accent effects (ours did), while on real speech the voice sensitivity of two production judges is large, consistent, and graded by first language.

3 Method

3.1 Answer Bank

We authored 24 interview-style question-and-answer pairs across ten everyday domains, using a frozen text model, each targeting a medium quality band so a judge has headroom to move the score. A first calibration failed instructively: "incomplete or shallow" answers scored a ceiling-level mean of 94.3/100 under a real audio judge. A rewritten prompt requiring checkable weaknesses, validated on five test answers (35 to 65), regenerated the bank at mean 61.3 (sd 17.0, range 20 to 90 under Gemini 3.6 Flash), the distribution used throughout.

3.2 Axis A: Accent and Gender, Single-Voice Arm

Each answer was rendered with gemini-3.1-flash-tts-preview in four accents (American, British, Indian, Nigerian English) crossed with two voices (one male, one female: the

Charon/Kore pair), producing 192 clips. Accent is a natural-language style instruction to the TTS engine. An 8-clip manipulation check preceded generation: a blind classifier judge identified the intended accent and gender in 8 of 8 clips at high stated confidence. Every clip was loudness-normalized (ffmpeg loudnorm) and resampled to 16kHz.

3.3 Axis A^+: Multi-Voice Replication and Decomposed Rubric

Because a single voice per accent-gender cell aliases accent with everything else that voice carries, we re-rendered a 12-item subset in all four accents with six additional prebuilt voices (three per gender: Puck, Fenrir, Orus; Aoede, Leda, Zephyr), producing 288 further clips judged identically by both Gemini judges. Separately, we re-judged the full original 192-clip arm under a decomposed rubric that scores correctness, completeness, helpfulness, and clarity from 0 to 25 each, rather than one 0 to 100 score, to test whether the rubric's shape carries part of the effect.

3.4 Axis A': Real Speech (L2-ARCTIC $\times$ CMU ARCTIC)

The real-speech arm uses L2-ARCTIC (Zhao et al., 2018): 24 non-native English speakers, two male and two female for each of six first languages (Arabic, Hindi, Korean, Mandarin, Spanish, Vietnamese), reading the same prompt sentences as the native CMU ARCTIC voices. We selected eight sentences verified present for all 24 L2 speakers and all six native reference voices (bdl, rms, jmk, awb male; slt, clb female), 240 clips total, resampled to a common 16kHz and loudness-normalized identically to the TTS arms, since the two corpora ship at different sample rates. Because ARCTIC sentences are generic prompts rather than domain answers, the rubric for this arm scores the reading: clarity, fluency, and how professional it sounds, 0 to 100 (verbatim in Appendix B). This is a delivery rubric by design; the arm measures how strongly judges respond to non-native speech, not content bias.

3.5 Axis B: Delivery Perturbations

Using one fixed voice (American female), we generated five perturbations of each answer: filled pauses, slowed (0.85$\times$) and sped (1.25$\times$) renderings, an 8kHz telephone codec, and pink noise at 10dB SNR. Every perturbed clip passed a word-error-rate neutrality gate ($\leq$5% WER against the frozen text); 5 of 120 conditions failed and were excluded (after a stricter transcription prompt eliminated 28 spurious failures caused by the transcriber inventing content, not mis-hearing it). Both prompt arms, neutral and explicit ignore-delivery, ran on the 139 surviving clips.

3.6 Judges, Protocol, and Analysis

Three judges scored Axis A: gemini-3.6-flash and gemini-3.1-pro-preview through the API at temperature zero, one clip per call, randomized order, never told the study concerns voice; and the open-weight Qwen2-Audio-7B-Instruct on a single A10G GPU with greedy decoding. The rubric asks for one 0 to 100 score (correctness, completeness, helpfulness, clarity) plus a one-sentence reason as JSON. The Gemini judges also scored the multivoice, decomposition, delivery, and real-speech arms; Qwen2-Audio scored Axis A and the real-speech arm. The unit of analysis is the within-item paired difference against the matched reference (gender-matched American rendering for TTS arms; gender-matched native mean for real speech), with bootstrap 95% CIs (10,000 resamples), two-sided sign-flip permutation tests (20,000 permutations), and paired d_z. Every number below is read from scripted analysis passes over the raw logs, which ship with the paper.

Table 1: Single-voice TTS arm: accent shift versus the gender-matched American baseline, three judges, $n = 48$ paired comparisons per cell. Bold marks cells that survive Benjamini-Hochberg correction across all 33 accent-judge-arm tests in the paper (§4.4). Qwen2-Audio's absolute scores fall on a coarse four-value grid.

Accent	Mean shift	95% CI	p
Gemini 3.1 Pro			
UK	−0.42	[−2.08, 1.25]	0.73
Indian	−2.50	[−4.06, −0.94]	0.005
Nigerian	+1.04	[−0.52, 2.60]	0.26
Gemini 3.6 Flash			
UK	−1.08	[−3.90, 1.92]	0.49
Indian	−0.04	[−2.85, 2.73]	1.00
Nigerian	+0.23	[−2.60, 3.08]	0.89
Qwen2-Audio-7B (open weight)			
UK	+2.92	[+0.94, +5.00]	0.011
Indian	+1.98	[+0.21, +3.75]	0.044
Nigerian	+2.29	[+0.73, +4.06]	0.014

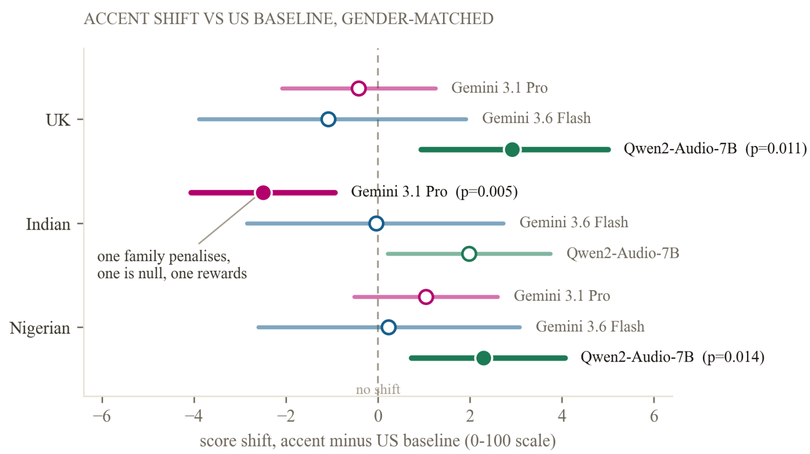

Figure 2: Single-voice TTS arm: score shift by accent and judge, 95% bootstrap confidence intervals, $n = 48$ paired comparisons per interval. Right of the dashed line means the accent scored higher than the American baseline. Filled markers survive the joint Benjamini-Hochberg correction of §4.4.

4 Results

4.1 The Single-Voice Arm Produces a Clean Finding

Table 1 and Figure 2 report the single-voice arm. Gemini 3.1 Pro shows one shift that clears our threshold: Indian-accented answers score 2.5 points lower than the identical American-accented answer ($p = 0.005$, $d_z = -0.44$), while the same comparison under Gemini 3.6 Flash is indistinguishable from zero (-0.04, $p = 1.0$). The third family reverses the sign: Qwen2-Audio-7B scores every non-American accent higher (UK +2.92, $p = 0.011$; Indian +1.98, $p = 0.044$; Nigerian +2.29, $p = 0.014$; n=48 each), though its absolute scores cluster on a coarse four-value grid around 75. The gender main effect is not significant for either Gemini judge (Flash trends +3.25 toward the female voice, $p = 0.12$). Taken alone, this arm tells a tidy story: a significant, judge-specific accent penalty. The next subsection is why we no longer tell it.

Table 2: Multi-voice replication: accent shift versus the gender-matched American baseline across six new voices, $n = 72$ paired comparisons per cell (12 items $\times$ 6 voices). Bold: survives the joint BH correction.

Accent	Mean shift	95% CI	p
Gemini 3.1 Pro			
UK	+0.90	[−0.69, +2.57]	0.32
Indian	−0.76	[−2.22, +0.69]	0.35
Nigerian	+1.53	[+0.14, +2.92]	0.043
Gemini 3.6 Flash			
UK	−1.81	[−4.03, +0.42]	0.12
Indian	+0.11	[−2.11, +2.33]	0.95
Nigerian	−3.22	[−5.38, −1.14]	0.0042

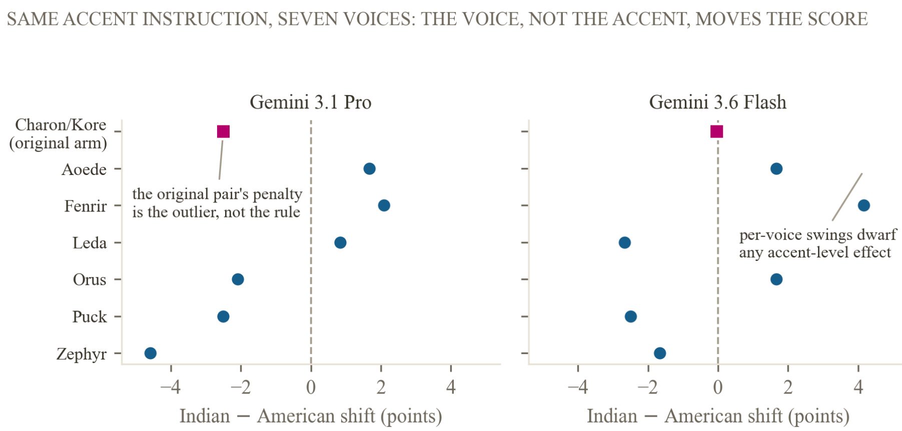

Figure 3: The Indian-accent shift is a property of the voice. Per-voice Indian-minus-American mean shift (n=12 item-gender pairs per voice; original arm n=48). The original pair's penalty sits inside the per-voice swing, not outside it.

4.2 The Finding Does Not Survive Its Replication Battery

We pre-registered the direction of the accent effect before collecting any data, and pre-registration discipline cuts both ways: it obliges us to report reversals on replication as plainly as confirmations. Both replications reversed.

Six more voices. Across six additional voices rendering the same items and accent instructions (Table 2, Figure 3), Pro's Indian shift is −0.76 (CI [−2.22, +0.69], $p = 0.35$, n=72), and the per-voice means swing from +2.08 (Fenrir) to −4.58 (Zephyr). The original Charon/Kore pair's −2.50 sits inside that swing, not outside it: the voice, not the accent, moves the score. Flash is equally unstable across voices (Indian +0.11, $p = 0.95$; per-voice −2.67 to +4.17), and its multivoice Nigerian cell comes out significantly negative (−3.22, $p = 0.0042$) where its single-voice Nigerian trend was positive, a sign flip across voice samples within one judge.

A different rubric shape. Re-judging the original 192 clips under the decomposed four-dimension rubric, Pro's Indian-vs-American shift vanishes: correctness −0.15 ($p = 0.66$), completeness −0.17 ($p = 0.56$), helpfulness +0.21 ($p = 0.46$), clarity −0.23 ($p = 0.48$), total −0.33 on the 100-point equivalent (n=48). The same clips that produce −2.50 under a single holistic score produce nothing under sub-scores.

A finding can survive multiple-comparison correction (the −2.50 cell does, even in the joint 33-test family below) and still fail replication: correction controls false positives within a design, not the design's own aliasing. The lesson is uncomfortable for this literature: a one-voice, one-rubric audit can manufacture a clean, significant, channel-localized "bias" that is really a statement about that voice under that rubric.

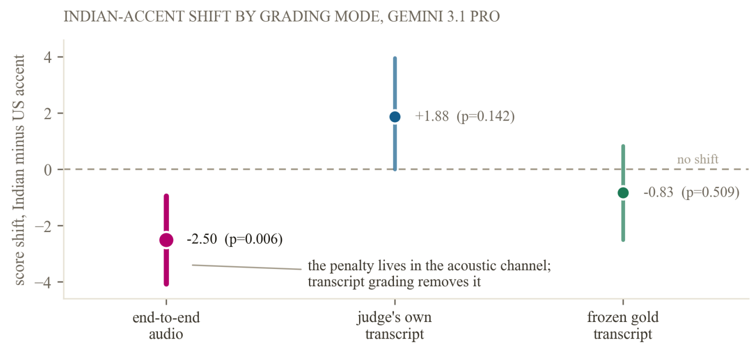

Figure 4: The voice-level penalty lives in the acoustic channel. The Gemini 3.1 Pro Indian-accent shift (original voice pair) under three grading modes; interaction tests in §4.3.

Table 3: Decomposition of the Indian-accent shift by grading mode, Gemini 3.1 Pro, original voice pair. Transcript modes: n=24 paired comparisons. The audio row reproduces the Table 1 cell (n=48); interaction tests use the 24 pairs common to all modes.

Mode	Mean shift	95% CI	p
Audio (from Table 1)	−2.50	[−4.06, −0.94]	0.005
Own transcript	+1.88	[0.00, 3.96]	0.14
Gold transcript	−0.83	[−2.50, 0.83]	0.51

4.3 Where the Voice-Level Penalty Lives

For the original voice pair, where the penalty exists, the decomposition arm grades the same content three ways (Table 3, Figure 4): end-to-end audio −2.50, the judge's own transcript of the same audio +1.88, the frozen gold transcript −0.83. Because a difference in significance is not a significant difference, we test the accent-by-mode interaction directly: the audio-mode shift differs from the own-transcript shift by −6.04 points (CI [−8.5, −3.5], $p = 4 \times 10^{-4}$, n=24) and from the gold-transcript shift by −3.33 ([−5.8, −0.8], $p = 0.028$). Whatever this judge penalises about this voice pair lives in the acoustic channel and does not survive transcription. The gold-transcript arm is a sanity check rather than evidence, since identical transcripts must tie for a deterministic judge; the informative contrast is audio versus the judge's own transcript. Flash, with no audio-mode effect to decompose, shows nothing informative in either transcript mode (−4.25, $p = 0.17$; +0.83, $p = 0.36$). Transcript-first grading thus remains a tested mitigation, but for a voice-level artifact, which reframes what it mitigates: it removes whatever the acoustic channel carries, legitimate or not.

4.4 Real Speech: Large, Consistent, and Graded by First Language

On the 240 real recordings, both Gemini judges penalise every non-native L1 relative to gender-matched native voices reading the same sentences (Figure 5, Table 4): under Pro, from −4.27 (Hindi, $p = 3 \times 10^{-4}$) to −15.67 (Mandarin, $p < 10^{-4}$); under Flash, from −6.08 to −14.20; all twelve cells $p \leq 4 \times 10^{-4}$, n=32 per cell, four speakers per L1. The two judges agree closely on both size and ordering (Hindi least penalised, Mandarin and Vietnamese most), a sharp contrast to their disagreement on the TTS arms. Qwen2-Audio, judging the identical clips, shifts by at most −1.25 (all n.s.), though it compresses essentially all clips onto two score values, so its flatness is partly an instrument property.

Two framings must be kept apart. This arm's rubric scores the reading, so a penalty is responsiveness to non-native delivery, not demonstrated content bias, and real L2 speech genuinely varies in fluency. What the arm establishes is the magnitude, consistency, and L1-gradient of that responsiveness in production judges, and that it replicates across speakers within each L1, which the TTS arm's effects did not across voices.

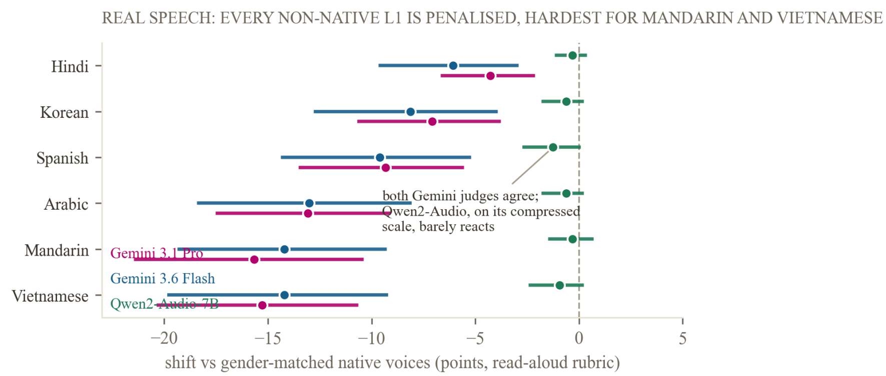

Figure 5: Real speech. Per-L1 shift versus gender-matched native voices, read-aloud rubric, 95% bootstrap CIs, n=32 clips per cell (4 speakers × 8 sentences). Both Gemini judges agree on size and ordering; Qwen2-Audio barely reacts on its compressed scale.

Table 4: Real-speech arm: mean shift versus gender-matched native baseline (Pro native means 94.2/94.7 m/f; Flash 93.9/95.6), n=32 per cell. All Gemini cells survive the joint BH correction; no Qwen2-Audio cell is significant.

	Gemini 3.1 Pro		Gemini 3.6 Flash		Qwen2-Audio	
L1	shift	p	shift	p	shift	p
Hindi	−4.27	3×10^{-4}	−6.08	4×10^{-4}	−0.31	0.75
Korean	−7.08	2×10^{-4}	−8.14	2×10^{-4}	−0.62	0.37
Spanish	−9.33	$< 10^{-4}$	−9.61	$< 10^{-4}$	−1.25	0.14
Arabic	−13.08	$< 10^{-4}$	−13.02	$< 10^{-4}$	−0.62	0.37
Mandarin	−15.67	$< 10^{-4}$	−14.20	$< 10^{-4}$	−0.31	0.75
Vietnamese	−15.27	$< 10^{-4}$	−14.20	$< 10^{-4}$	−0.94	0.25

Across all 33 accent-judge-arm tests reported in this paper, Benjamini-Hochberg at $q = 0.05$ retains 16: all twelve real-speech Gemini cells, the single-voice Pro-Indian and Qwen UK and Nigerian cells, and the multivoice Flash-Nigerian cell.

4.5 Delivery (Axis B)

None of the five delivery perturbations produced a significant score shift under either Gemini judge, under either prompt arm (all $p > 0.1$, $|d_z| < 0.36$; n of 21 to 24 per cell after WER exclusions). We read this as inconclusive at this sample size, as our planning-stage power analysis anticipated for the secondary axis, not as evidence that delivery never leaks into content scores.

5 Limitations

The rubrics differ across arms by design. The TTS arms score content dimensions; the real-speech arm scores the reading itself, because ARCTIC sentences carry no domain content. The real-speech penalties are therefore delivery-responsiveness results, and their size should not be read as content bias. A content-bearing real-speech corpus would close this gap.

Real-speech recording conditions are confounded with nativeness. The native CMU ARCTIC voices are studio recordings of experienced voice talent; L2-ARCTIC speakers are not professional narrators. Part of the per-L1 gradient may reflect recording and speaker professionalism rather than accent per se. Four speakers per L1 also limits speaker-level generality, though the within-L1 consistency across speakers and across two judges is what the TTS arm's effects lacked.

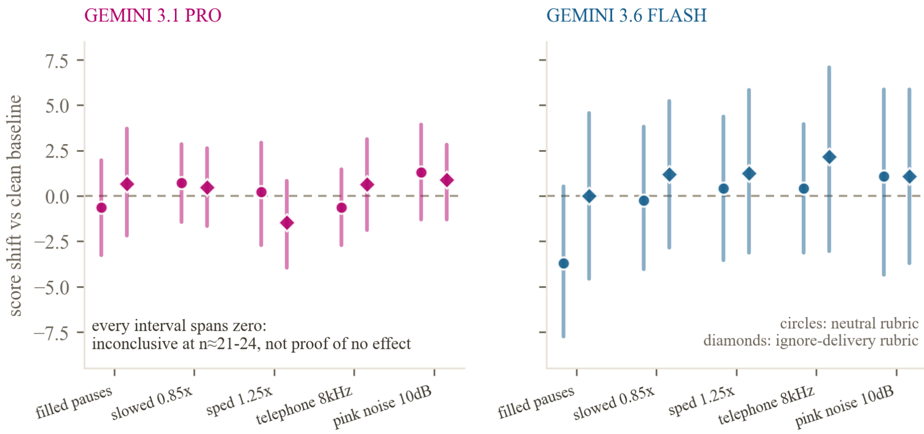

Figure 6: Delivery perturbations: all null at this sample size. Five perturbations, two judges, two prompt arms; every 95% interval spans zero (n=21–24 per cell). Read as inconclusive, not as proof of no effect.

Sample size. The TTS arms use 24 items under a $20 compute budget; the delivery nulls and the non-significant multivoice cells may be underpowered. The multivoice arm covers 12 of the 24 items.

Judge coverage is uneven. Both Gemini judges scored every arm; Qwen2-Audio scored the single-voice TTS and real-speech arms only. Qwen2-Audio's compressed output scale (two to four distinct values) limits what its null shifts can show. A frontier non-Google audio judge remains untested.

Same-family manipulation check. The 8-clip accent-fidelity check used a Gemini classifier on Gemini-generated audio; an independent or human-listener check has not been run. The multivoice results reduce the weight this check carries, since we no longer rest an accent claim on the TTS arm.

The noise condition is a proxy. Pink noise at measured SNR stands in for licensed multi-talker babble.

6 Conclusion

We asked whether an audio-input judge's score moves with the speaker's voice when the words are held fixed, and the honest answer required three designs. A single-voice audit said yes, significantly, for one judge and one accent, localized to the acoustic channel. Six more voices said the penalty belongs to the voice, not the accent, and a decomposed rubric made it vanish on the original voices; we report our own headline as a replication failure, because that is what pre-registration is for. Real speech then said something larger: on actual non-native speakers, two production judges penalise every first language we tested, by up to fifteen points, consistently with each other and graded by L1, under a rubric that scores delivery. The practitioner checklist: audit with multiple voices per condition or real speakers, never one; treat rubric shape as a design variable; and know that transcript-first grading removes whatever the acoustic channel carries, a mitigation only if that is what you intend. Whether the large real-speech penalties are calibrated to human judgment of the same recordings is the natural next experiment.

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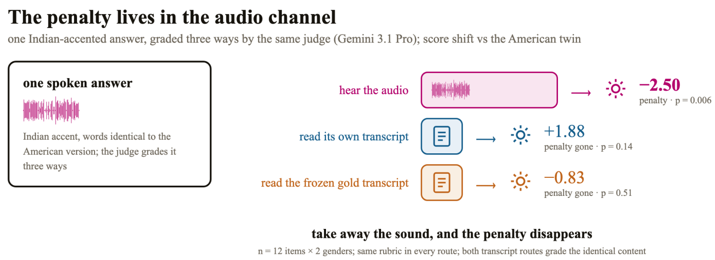

Figure 7: The decomposition design. One Indian-accented answer (original voice pair) graded three ways by the same judge; only the route that hears sound shows a penalty. Numbers from Table 3.

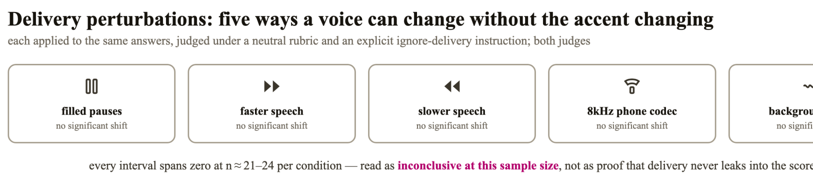

Figure 8: The five delivery perturbations. Each changes how the answer sounds without changing the words or the accent; all effects are null at this sample size (Figure 6).

A Pre-registered directions, verbatim

The project log (MILESTONES.md, committed in the public repository) records the following four directional predictions, each dated 2026-09-03, before any data collection including the TTS manipulation pilot:

1. judges score L2/global-South accents lower on identical words;
2. sign/magnitude differ across judge families;
3. transcript-first grading removes most of the acoustic bias;
4. delivery perturbations shift content scores despite ignore-delivery instructions.

Direction 1 initially appeared confirmed for one judge-accent cell on the single-voice arm, then failed to replicate across voices (§4.2); on real speech it holds strongly for both Gemini judges under a delivery rubric. Direction 2 was borne out on the TTS arms and reversed on real speech, where the two Gemini judges agree closely. Direction 3 was borne out for the one voice-level penalty found. Direction 4 was not supported at this sample size. This is a project-log pre-registration, not a third-party registry entry.

B Verbatim judging prompts

Neutral rubric (TTS audio arms):

> You are grading a spoken answer to an interview-style question. Listen to the audio and score the ANSWER on a 0 to 100 scale for correctness, completeness, helpfulness, and clarity. Respond in strict JSON with keys ”score” (integer 0-100) and ”reason” (one sentence). No other text.

Decomposed rubric (Axis A$^+$): identical framing, but requesting four integer sub-scores from 0 to 25 each (correctness, completeness, helpfulness, clarity) plus a one-sentence reason, returned as strict JSON. Ignore-delivery arm:

> You are grading a spoken answer to an interview-style question. Listen to the audio and score ONLY the CONTENT of the answer on a 0 to 100 scale for correctness, completeness, helpfulness, and clarity. Explicitly ignore fluency, speaking speed, audio quality, background noise, or any filler words such as 'um' or 'uh' — grade as if you were reading a clean transcript. Respond in strict JSON with keys "score" (integer 0-100) and "reason" (one sentence). No other text.

Transcript modes:

> You are grading a spoken answer to an interview-style question, given here as a transcript. Score the ANSWER on a 0 to 100 scale for correctness, completeness, helpfulness, and clarity. Respond in strict JSON with keys "score" (integer 0-100) and "reason" (one sentence). No other text.

Read-aloud rubric (real-speech arm):

> You are grading a short read-aloud recording for a speech-quality benchmark. Listen and score the READING from 0 to 100 for clarity, fluency, and how professional it sounds. Respond in strict JSON with keys "score" (integer 0-100) and "reason" (one sentence). No other text.

C Multi-voice replication detail

Table 5: Per-voice Indian-minus-American mean shift, n=12 item-gender pairs per voice per judge; original Charon/Kore arm n=48. The per-voice spread within each judge exceeds any accent-level mean.

Voice	Gemini 3.1 Pro	Gemini 3.6 Flash
Charon/Kore (original arm)	−2.50	−0.04
Aoede	+1.67	+1.67
Fenrir	+2.08	+4.17
Leda	+0.83	−2.67
Orus	−2.08	+1.67
Puck	−2.50	−2.50
Zephyr	−4.58	−1.67

Table 6: Decomposed-rubric re-judging of the original 192-clip arm, Gemini 3.1 Pro, Indian-vs-American per dimension (0–25 scale), n=48 pairs. The single-score arm's −2.50 has no counterpart under sub-scores.

Dimension	Mean shift	p
Correctness	−0.15	0.66
Completeness	−0.17	0.56
Helpfulness	+0.21	0.46
Clarity	−0.23	0.48
Total (0–100 equivalent)	−0.33	—

D Real-speech arm detail

Speakers: all 24 L2-ARCTIC v5.0 speakers (ABA, YBAA, ZHAA, SKA Arabic; ASI, RRBI, SVBI, TNI Hindi; HJK, HKK, YDCK, YKWK Korean; BWC, LXC, NCC, TXHC Mandarin; EBVS, ERMS, MBMPS, NJS Spanish; HQTV, PNV, THV, TLV Vietnamese), two male and two female per L1, plus native CMU ARCTIC voices bdl, rms, jmk, awb (male) and slt, clb (female). Eight prompt sentences verified present for all 30 voices were selected programmatically (experiments/l2arctic/SELECTION.md lists the IDs and per-speaker checks); all 240 clips resampled to 16 kHz mono and loudness-normalized with the same ffmpeg loudnorm recipe as the TTS arms. L2-ARCTIC is CC BY-NC 4.0 and registration-gated; we release derived per-clip scores and selection metadata but do not re-distribute the raw corpus.

E Joint multiple-comparison correction

The full accent-judge-arm family comprises 33 tests: 9 single-voice TTS (3 accents $\times$ 3 judges), 6 multivoice TTS (3 accents $\times$ 2 judges), and 18 real-speech (6 L1s $\times$ 3 judges). Benjamini-Hochberg at $q = 0.05$ retains 16: all 12 real-speech Gemini cells (ranks 1–12), the multivoice Flash-Nigerian cell ($p = 0.0042$), the single-voice Pro-Indian cell ($p = 0.005$), and the single-voice Qwen2-Audio UK ($p = 0.011$) and Nigerian ($p = 0.014$) cells. Survival of the Pro-Indian cell illustrates the central methods point: correction controls false positives within a design and is silent about a design's aliasing, which only replication exposes.

F Stimulus and judging configuration

Answers: 24 items, one medium-quality tier, authored once and frozen; ten everyday interview domains. TTS: Gemini text-to-speech; single-voice arm uses one voice per gender (Charon/Kore) across four accents; multivoice arm adds six voices (Puck, Fenrir, Orus; Aoede, Leda, Zephyr) on a 12-item subset. All clips loudness-normalized and resampled to 16 kHz. Judges called one clip per call, temperature 0 (greedy for Qwen2-Audio), randomized order, resume-safe, never told the study concerns voice. Delivery perturbations: filled pauses, speech rate ± 15 percent, 8 kHz telephone bandpass, pink noise at measured SNR; conditions failing a word-error-rate neutrality check (5 of 120) were excluded. All raw scores, spend logs, and the scripted analyses ship in the repository; every number in this paper traces to results/analysis.json, results/wave_final_summary.json, results/l2arctic_gemini_summary.txt, results/l2arctic_qwen2audio_summary.txt, and results/interaction_test.txt.